

**CKDL**

Croydon Kowloon Designs Ltd.

GarmentFlow

Production Order Management System — Executive Summary

DOCUMENT DATE

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VERSION

1.0

CLASSIFICATION

Internal — Confidential

PREPARED BY

CKDL Information Technology

1 Project Objective

GarmentFlow is a purpose-built, web-based Production Order Management System developed exclusively for Croydon Kowloon Designs Ltd. (CKDL) to digitise and centralise the end-to-end garment manufacturing workflow.

The system replaces manual, spreadsheet-driven processes with a structured, role-controlled digital platform that provides real-time visibility across the full order lifecycle — from initial customer order receipt through production scheduling to final shipment and invoicing.

Key objectives of the system include:

- › Eliminate data duplication and manual reconciliation between departments
- › Enforce accountability through a structured, group-based access control framework
- › Provide management with consolidated, real-time visibility of factory operations
- › Automate sequential status transitions across orders, production runs, and shipments

- › Maintain a single source of truth for customer, brand, style, and production line master data
- › Ensure auditability through timestamped records and system-generated reference numbers

2 Admin Panel — Functionality

The Admin Panel is exclusively accessible to designated administrator accounts via a dedicated, segregated login portal. Administrator credentials are rejected at the general user login and vice versa, ensuring complete separation of the administrative interface from day-to-day operational use.

MASTER DATA MANAGEMENT

- › **Customers** — Maintain buyer profiles including company name, contact details, country, payment terms (Net 30, Letter of Credit, TT Advance), and active/inactive status. Each customer may be linked to multiple brands.
- › **Brands** — Register buyer brands with unique code, description, and country of origin. Brands are linked to customers through a flexible many-to-many relationship.
- › **Garment Styles** — Define full style specifications: style number, category (Shirt, Trouser, Jacket, T-Shirt, etc.), fabric type and composition, season code (e.g. SS26, AW26), gender, Standard Allowed Minutes (SAM), standard cost per unit, and buyer style reference. Each style is linked to a brand and optionally to a specific customer.
- › **Production Lines** — Register factory production lines with line code, daily capacity (units/day), assigned supervisor, and operational status (Active / Maintenance / Inactive).

USER & ACCESS ADMINISTRATION

- › **User Management** — Create, edit, and deactivate system user accounts. Each user is classified as Administrator or General User. Administrators are restricted to the Admin Panel; General Users access only the transaction modules. Self-deletion by an active administrator is prevented by the system.
- › **Groups & Permissions** — Define named permission groups with granular, module-level access flags:

Can Access Orders

Can Access Productions

Can Access Shipments

Each general user is assigned to one or more groups. A user's access is determined by the cumulative permissions across all their active groups.

ADMIN DASHBOARD

Provides a real-time system overview displaying total counts for: Users, Permission Groups, Customers, Brands, Garment Styles, and Production Lines. Quick-action shortcuts allow administrators to create new master data records or manage users and groups in a single click.

3 User Panel — Functionality

General users log in through the standard login portal and are directed to a personalised dashboard. Their accessible modules are governed entirely by the permission groups assigned by the administrator. The sidebar navigation displays all transaction modules, with access enforced at the route level.

ORDER RECEIPT MODULE

- › Create new customer orders with auto-generated, sequential reference numbers (**ORD-YYYYMM####**)
- › Record buyer Purchase Order (PO) number, garment style, order quantity, unit price, and currency (USD, EUR, GBP, AUD, CAD, BDT)
- › Capture order date and required delivery date with overdue highlighting
- › Search and filter orders by customer, status, order number, or PO number
- › View detailed order records with linked production runs and shipment invoices

Order lifecycle status progression:

Pending →

Confirmed →

In Production →

Ready to Ship →

Shipped →

Cancelled

PRODUCTION MODULE

- › Schedule production runs against confirmed orders with auto-generated numbers (**PRD-YYYYMM####**)
- › Assign orders to specific production lines; record planned and actual start/end dates

- › Track output quantity (passed) and rejected quantity per run
- › System automatically calculates production efficiency (%) from planned vs. actual output
- › Completing a production run automatically advances the linked order to *"Ready to Ship"*

Production run status progression:

Scheduled →

In Progress →

Completed →

On Hold →

Cancelled

SHIPMENT & INVOICE MODULE

- › Create shipment invoices against orders with auto-generated numbers (INV-YYYYMM####)
- › Record shipped quantity, carton count, net weight, and gross weight (KG)
- › Capture full shipping details: port of loading, port of discharge, vessel name, Bill of Lading (B/L) number, ETD, and ETA
- › Record invoice value (unit price × shipped quantity) in the applicable currency
- › Creating a shipment automatically advances the linked order status to *"Shipped"*

Shipment status progression:

Draft →

Confirmed →

Shipped →

Delivered

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Database Relationships

The system uses a fully relational PostgreSQL database. All tables are managed through SQLAlchemy ORM with defined foreign keys, cascade rules, and eager-loaded relationships. The schema is structured around the core order lifecycle, with master data entities feeding into transaction entities.

ENTITY RELATIONSHIP	TYPE	DESCRIPTION
Customer ↔ Brand	M-M via CustomerBrand	A buyer may carry multiple brands; a brand may be shared across multiple buyers.
Customer → Orders	1-M	Each order is raised against a single customer. A customer may have many orders over time.

ENTITY RELATIONSHIP	TYPE	DESCRIPTION
GarmentStyle → Orders	1-M	Each order references a single garment style. A style may appear across multiple orders.
Brand → GarmentStyles	1-M	A brand owns multiple garment styles. Each style is linked to one brand.
Order → Productions	1-M	A single order can have multiple production runs — across different lines or time periods.
Order → Shipments	1-M	A single order can have multiple partial or full shipment invoices raised against it.
ProductionLine → Productions	1-M	Each production run is assigned to one production line. A line may run many batches.
User ↔ Group	M-M via UserGroup	A user may belong to multiple permission groups. A group may contain many users. Access is determined by the union of group-level permission flags.

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Technology Stack

LAYER	TECHNOLOGY & DETAILS
Backend Framework	FastAPI (Python 3.12) — asynchronous, high-performance web framework with automatic OpenAPI documentation
Database	PostgreSQL — open-source, ACID-compliant relational database management system
ORM & Driver	SQLAlchemy 2.0 (async mode) with asyncpg driver — non-blocking database operations for high concurrency

LAYER	TECHNOLOGY & DETAILS
Authentication	JWT (JSON Web Tokens) — HS256 algorithm, delivered as HttpOnly cookies; 8-hour session expiry
Password Security	bcrypt — industry-standard adaptive hashing algorithm with salt rounds
Templating Engine	Jinja2 — server-side HTML rendering with shared global context and template inheritance
Frontend	Custom CSS Design System — Apple-inspired, fully responsive; no frontend framework dependency
Web Server	Uvicorn (ASGI) — lightning-fast asynchronous server gateway interface
Configuration	Pydantic Settings — typed, validated configuration with .env file support
Middleware	Starlette BaseHTTPMiddleware — request-level access control enforcement before route handlers
Database Migrations	SQLAlchemy auto-create on startup + versioned manual SQL migration scripts

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Security Architecture

 **Dual Login Portals**

Admin and general user logins are completely separated. Admin credentials are rejected at the general portal and vice versa, preventing privilege escalation.

 **Middleware Access Control**

Every HTTP request is intercepted by AccessControlMiddleware before reaching route handlers. Unauthenticated requests are instantly redirected to the appropriate login page.

 **HttpOnly JWT Cookies**

Authentication tokens are stored in HttpOnly cookies, making them inaccessible via

 **Group-Based Permissions**

Module access is controlled through permission group assignments. Adding or removing a user

JavaScript and protecting against XSS-based token theft. SameSite=Lax mitigates CSRF risk.

from a group instantly changes their effective access — no code changes required.

Route-Level Protection

Admin routes (/admin, /customers, /brands, /styles, /lines) require administrator status. Transaction routes require an active authenticated session with a valid JWT.

Self-Delete Prevention

Administrators cannot delete their own account through the user management interface, preventing accidental lockout from the administration panel.

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This document is prepared for internal use only and is strictly confidential.

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